

Hydroelasticity with Gravity: A Frequency Response Function Approach

Comparative Study of Single and Full Tank Systems (Incompressible Formulation)

Khosrov Jabrayilov¹, Mumen Wehbe¹

^aMaster, Conservatoire National des Arts et Métiers (Cnam), Paris, France

Abstract

This project presents a comprehensive numerical investigation into the hydroelastic behavior of fluid storage systems, assuming an incompressible fluid model. The study examines the impact of dual-sided fluid loading and gravity effects on structural dynamics. The methodology progresses from the validation of a *Single Tank* model to a complex *Full Tank* ("sandwich") configuration. A custom Finite Element code was developed in MATLAB to solve the coupled (u, p) formulation. A major focus of the research was the resolution of critical numerical instabilities. In the fluid domain, spurious hourglassing modes were eliminated by implementing a high-order Gauss quadrature (3×3) . Simultaneously, a severe shear locking phenomenon was identified in the thin structural beam, initially resulting in a massive 1356% error in the fundamental frequency (103 Hz vs 7 Hz). This was resolved by implementing a Selective Reduced Integration (SRI) scheme, where the material stiffness matrix is decomposed to under-integrate shear terms (1×1) while fully integrating normal stresses (2×2) . The final results demonstrate that the "sandwich" configuration significantly amplifies added mass effects, reducing the fundamental frequency by 70%, while the inclusion of gravity is proven essential for capturing low-frequency sloshing modes (< 2 Hz).

Keywords: Fluid-Structure Interaction, Incompressible Hydroelasticity, Sloshing, Finite Element Method, Frequency Response Function

1. Introduction

Fluid-Structure Interaction (FSI), and more specifically *hydroelasticity*, constitutes a critical design consideration for various industrial applications, ranging from aerospace fuel tanks and nuclear fuel assemblies to LNG carriers. In these systems, flexible structural components are frequently subjected to fluid loading on multiple boundaries. This coupling creates complex dynamic behaviors where the inertia of the fluid significantly alters the structural response, necessitating advanced numerical modeling techniques beyond standard solid mechanics.

This project, conducted by Master students of the "Structural Mechanics and Coupled Systems" program at the Conservatoire National des Arts et Métiers, investigates the numerical modeling of such systems using an *incompressible fluid assumption*. The study was carried out during the Fall Semester (December 2025 – January 2026). The primary scientific motivation is to quantify the "sandwich effect"—where a structure is bounded by fluid on two sides—and to determine the specific contribution of gravity terms in the mathematical formulation. Unlike acoustic problems where fluid compressibility plays a role, this study focuses on the hydrodynamic added mass and gravity-driven sloshing phenomena relevant to low-frequency excitations.

* Supervisor: Christophe Hoareau

1.1. Problem Statement and Objectives

The central challenge of this project is to predict the Frequency Response Function (FRF) for a flexible beam positioned between two fluid domains. This requires coupling elastodynamic solid equations with the fluid's Laplace equations while accounting for gravity-driven free-surface conditions. To address this complexity, the study adopts a progressive methodology that begins by isolating the fluid dynamics to compute pure sloshing modes within rigid walls, thereby validating the gravity stiffness matrix. Following this, the structure is analyzed in a vacuum to establish a baseline elastic frequency, which allows for a direct comparison when the fluid is introduced. The investigation then proceeds to a hydrodynamic coupling scenario that neglects gravity, isolating the pure added mass effect of the fluid on the structure. The final stage involves the complete hydroelastic problem, incorporating gravity to fully characterize the interaction between the structural vibrations and the fluid sloshing modes.

1.2. Project Planning (Gantt Chart)

The project was executed between December 8, 2025, and January 19, 2026. The workflow followed the sequential implementation of the Finite Element code, validation of the single-tank case, and final analysis of the full-tank system, as detailed in Figure 1.

Figure 1: Project realization timeline (Dec 2025 - Jan 2026).

Task Breakdown:

- Week 1 (Dec 8-14): Code Architecture & Single Tank Mesh Gen.
- Week 2 (Dec 15-21): Matrix Assembly (KM, HQ) & Dry Analysis.
- Week 3 (Dec 22-28): Gravity Implementation (S Matrix) & Sloshing Modes.
- Week 4 (Dec 29-Jan 4): Coupling Matrix (C) & Hydrodynamic FRF.
- Week 5 (Jan 5-11): Full Tank Expansion & Integration/Hourglass Fix.
- Week 6 (Jan 12-19): Final Results, Comparisons & Report Writing.

2. Mathematical Formulation and Discretization

The physical system consists of a structural domain Ω_s (flexible beam) and a fluid domain Ω_f (water reservoirs). The domains interact at the interface Σ and are bounded at the top by a free surface Γ . The problem is modeled using a mixed (u, p) formulation, where u is the structural displacement and p is the fluid pressure. The geometric definitions are illustrated in Figure 2.

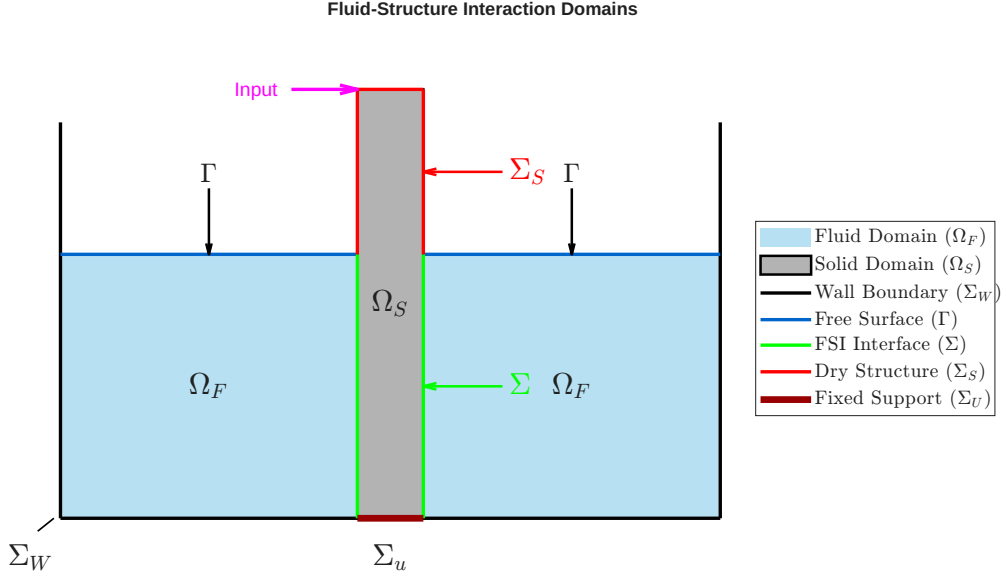


Figure 2: 2D Computational domain definition showing the Fluid (Ω_F) and Solid (Ω_S) domains, the fluid-structure interface (Σ), and the free surface (Γ).

2.1. Hypotheses and Governing Equations

The mathematical model is derived under the following physical assumptions:

- **Structure:** Linear elastic, isotropic, and homogeneous material undergoing small perturbations.
- **Fluid:** Inviscid, irrotational, and incompressible ($c \rightarrow \infty$).
- **Gravity:** Considered only on the free surface boundary (linearized wave theory).

The solid dynamics in Ω_S are governed by the elastodynamic equation. Assuming harmonic excitation $u(t) = \tilde{u}e^{j\omega t}$:

$$\nabla \cdot \boldsymbol{\sigma}(u) + f_v = -\omega^2 \rho_S u \quad \text{in } \Omega_S \quad (1)$$

The fluid in Ω_F is governed by the Laplace equation, derived from the conservation of mass and momentum for an incompressible fluid:

$$\Delta p = 0 \quad \text{in } \Omega_F \quad (2)$$

2.2. Boundary Conditions and Gravity Effects

The behavior of the free surface Γ depends critically on the inclusion of gravity. We distinguish between two formulations used in this study:

2.2.1. Case A: Full Hydroelasticity (With Gravity)

When gravity g is included, it acts as a restoring force for surface waves (sloshing). The linearized Bernoulli condition applies:

$$\frac{\partial p}{\partial n} = \frac{\omega^2}{g} p \quad \text{on } \Gamma \quad (3)$$

This condition introduces the "Sloshing Matrix" term in the weak form, essential for capturing low-frequency surface modes.

2.2.2. Case B: Zero-Gravity Limit ("Acoustic" Approximation)

In the limit where gravity is neglected ($g \rightarrow 0$), the hydrostatic restoring force vanishes. The boundary condition simplifies to a homogeneous Dirichlet condition:

$$p = 0 \quad \text{on } \Gamma \quad (4)$$

In this configuration, sloshing modes disappear ($\omega \rightarrow 0$), and the fluid acts purely as an added mass.

2.3. Weak Formulation

To solve the system numerically, we apply the weighted residual method. Multiplying the strong forms by test functions δu (displacement) and δp (pressure) and applying Green's theorem yields the integral weak forms.

Fluid Weak Form:

$$\int_{\Omega_F} \frac{1}{\rho_F} \nabla p \cdot \nabla \delta p \, d\Omega - \omega^2 \int_{\Gamma} \frac{1}{g\rho_F} p \delta p \, d\Gamma - \omega^2 \int_{\Sigma} u \cdot n \delta p \, d\Gamma = 0 \quad (5)$$

Structural Weak Form: The structural equation includes the work done by fluid pressure forces acting on the interface Σ :

$$\int_{\Omega_S} \sigma(u) : \epsilon(\delta u) \, d\Omega - \omega^2 \int_{\Omega_S} \rho_S u \cdot \delta u \, d\Omega - \int_{\Sigma} p n \cdot \delta u \, d\Gamma = 0 \quad (6)$$

2.4. Finite Element Discretization

Using shape functions N_u and N_p to approximate the fields u and p , the integral forms are transformed into a linear matrix system. The elementary matrices are defined as:

$$H = \int_{\Omega_F} \frac{1}{\rho_F} \nabla N_p^T \nabla N_p \, d\Omega \quad (\text{Fluid Stiffness}) \quad (7)$$

$$S = \int_{\Gamma} \frac{1}{g\rho_F} N_p^T N_p \, d\Gamma \quad (\text{Sloshing/Gravity}) \quad (8)$$

$$C = \int_{\Sigma} N_u^T \mathbf{n} N_p \, d\Gamma \quad (\text{Fluid-Structure Coupling}) \quad (9)$$

This leads to the following global coupled system:

$$\left(\begin{bmatrix} K_S & -C \\ 0 & H \end{bmatrix} - \omega^2 \begin{bmatrix} M_S & 0 \\ \rho_F C^T & \mathbf{M}_{\text{fluid}} \end{bmatrix} \right) \begin{Bmatrix} U \\ P \end{Bmatrix} = \begin{Bmatrix} F \\ 0 \end{Bmatrix} \quad (10)$$

where $\mathbf{M}_{\text{fluid}}$ is $\frac{1}{g}S$ for the full hydroelastic case, and $\mathbf{0}$ for the zero-gravity limit.

3. Numerical Implementation

The computational framework was developed in a modular environment, separating the mesh generation, element-level computations, and global system assembly into distinct algorithmic procedures.

3.1. Model Parameters and Simulation Setup

To ensure reproducibility, the geometric and material properties used for all simulations are summarized in Table 1. The structural parameters correspond to a thin steel beam, while the fluid properties represent water at standard conditions.

Table 1: Physical and Geometric Parameters used in the Finite Element Model.

Domain	Parameter	Symbol	Value
Fluid	Density	ρ_F	1000 kg/m ³
	Speed of Sound	c_0	1438 m/s
	Gravity	g	9.81 m/s ²
	Tank Width	L_x	0.231 m
	Fluid Depth	L_{yf}	0.210 m
Structure	Young's Modulus	E	210 GPa
	Poisson's Ratio	ν	0.3
	Density	ρ_S	7800 kg/m ³
	Beam Thickness	L_{xs}	0.57 mm
	Beam Height	L_{ys}	0.266 m
	Damping Ratio	ζ	2%
Mesh	Fluid Grid	$N_x \times N_y$	16 × 15 elements
	Structure Grid	$N_x \times N_y$	4 × 19 elements

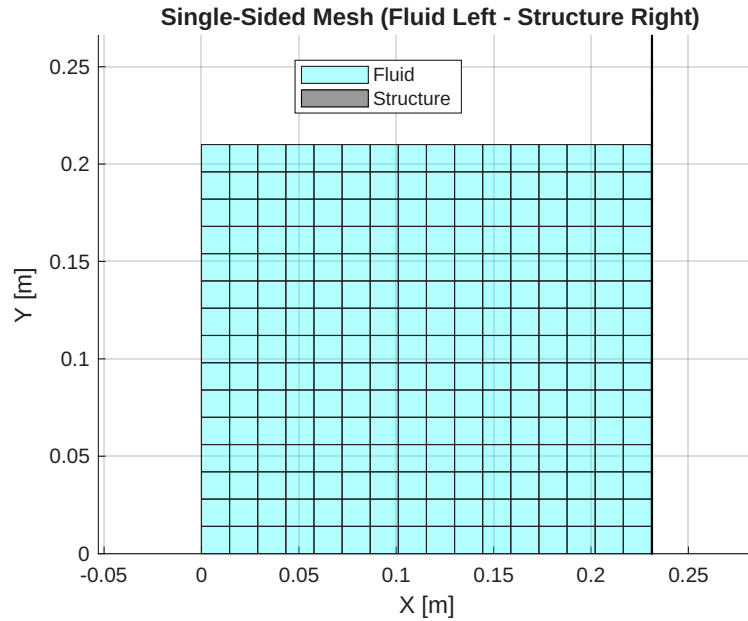


Figure 3: Single-Sided Mesh (Fluid Left - Structure Right). This visualization shows the discretization corresponding to the parameters defined in Table 1 (16 × 15 fluid elements).

3.2. Global Matrix Assembly Algorithms

The transition from local element physics to the global system response is handled by specific assembly routines implementing the direct stiffness method.

3.2.1. Structural Assembly (DOF Mapping)

The routine `KM_global.m` constructs the global structural stiffness (K_S) and mass (M_S) matrices. The structure is modeled with 2 degrees of freedom (DOF) per node (u_x, u_y). The algorithm iterates over every element e , retrieving the node indices $\mathbf{n}_e = [n_1, n_2, n_3, n_4]$.

To assemble the system correctly, the routine explicitly maps these local node indices to the global DOF vector using the following relations:

$$\text{DOF}_{2k-1}^{(e)} = 2n_k - 1 \quad (\text{x-displacement}) \quad (11)$$

$$\text{DOF}_{2k}^{(e)} = 2n_k \quad (\text{y-displacement}) \quad (12)$$

This generates an elemental index vector $\mathbf{I}_e$ of size 8×1 . The global accumulation is then performed:

$$\mathbf{K}_{global}(\mathbf{I}_e, \mathbf{I}_e) \leftarrow \mathbf{K}_{global}(\mathbf{I}_e, \mathbf{I}_e) + \mathbf{K}_{local}^{(e)} \quad (13)$$

3.2.2. Sloshing Matrix Assembly

The routine `S_global.m` handles the gravity stiffness matrix. Unlike the domain integrals, this function iterates only over the subset of 1D line elements on the free surface Γ .

Since the fluid pressure is a scalar field (1 DOF per node), the mapping is direct ($\text{DOF}_k = n_k$). For each surface element e with nodes $\mathbf{J}_e$, the local contribution $\mathbf{S}_{local}$ is integrated into the global sparse matrix:

$$\mathbf{S}_{global}(\mathbf{J}_e, \mathbf{J}_e) \leftarrow \mathbf{S}_{global}(\mathbf{J}_e, \mathbf{J}_e) + \mathbf{S}_{local}^{(e)} \quad (14)$$

3.3. Eigenvalue Solvers and Sloshing Analysis

Once the global matrices are assembled, the code utilizes MATLAB's sparse eigenvalue solver (`eigs`) to compute the natural frequencies and mode shapes. To validate the implementation, the code first solves for pure sloshing modes in a rigid tank, reducing to the generalized eigenvalue problem:

$$(H - \omega^2 \frac{1}{g} S) \{\phi\} = 0 \quad (15)$$

3.4. Space Integration and Stability

A critical aspect of numerical stability was the selection of the integration scheme. Initial tests using reduced integration for the quadrilateral elements resulted in spurious "hourglass" modes. To resolve this, the fluid domain elements are evaluated using a full 3×3 Gauss quadrature rule.

4. Results and Analysis

The numerical results are presented progressively, following a rigorous validation strategy designed to isolate and verify each physical mechanism before attempting the fully coupled solution.

4.1. Single Tank Validation

Before addressing the complexities of the full "sandwich" system, a preliminary validation was performed on a simplified single-tank configuration. This step verified the correct implementation of the free-surface boundary condition and the gravity stiffness matrix (S) in isolation.

4.1.1. Rigid Tank Sloshing: Validation Case

To validate the fluid formulation, we analyze a rigid tank ($u = 0$) where the problem reduces to finding the pressure field p satisfying $\Delta p = 0$ with $\partial_n p = (\omega^2/g)p$ on the free surface.

Analytical Solution: The results are compared against the wave theory dispersion relation:

$$\omega_n^2 = gk_n \tanh(k_n h), \quad k_n = \frac{n\pi}{L} \quad (16)$$

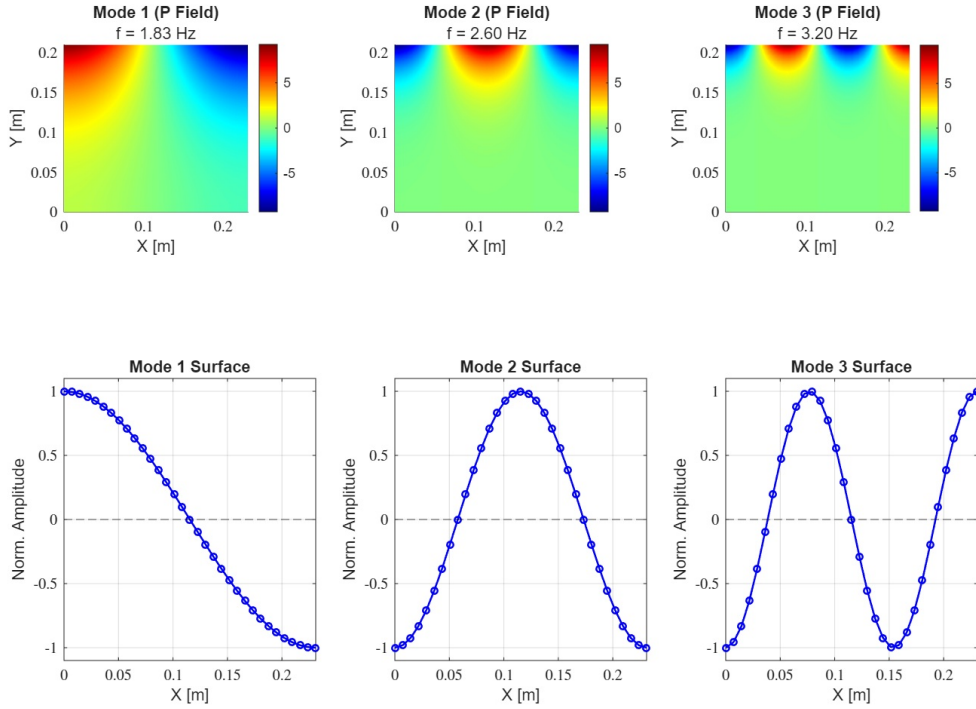


Figure 4: Sloshing Modes 1-3. The top row displays the free surface elevation (η). The bottom row shows the corresponding pressure field (P).

Modes Analysis:

- **Mode 1 (1.83 Hz):** Fundamental antisymmetric mode; fluid sloshes left to right.
- **Mode 2 (2.62 Hz):** First symmetric mode.
- **Mode 3 (3.20 Hz):** Second antisymmetric mode.

4.1.2. Dry Structure Verification & Shear Locking Resolution

A critical verification step was performed on the dry structural beam to benchmark the elastodynamic matrices (K_S , M_S) against analytical beam theory. The theoretical fundamental frequency for a cantilever beam is given by:

$$f_{ana} = \frac{1.8751^2}{2\pi L^2} \sqrt{\frac{EI}{\rho A}} \approx 7.08 \text{ Hz} \quad (17)$$

Initial simulations using standard fully integrated linear quadrilateral elements (Q4) yielded a massive discrepancy (103.08 Hz). This 1356% error is a classic manifestation of *Shear Locking*.

Physical Mechanism: Parasitic Shear. Standard Q4 elements with full Gauss integration (2×2) suffer from a numerical pathology known as "parasitic shear." When a linear element undergoes pure bending, the edges remain straight, forcing the internal angles to change. The element interprets this angle change as a physical shear strain ($\gamma_{xy} \neq 0$), even though pure bending should be shear-free. This spurious strain generates substantial artificial strain energy, making the numerical model excessively stiff. As the aspect ratio of the element increases (i.e., for thin beams), this parasitic energy dominates the solution, leading to the "locked" result of 103 Hz.

Resolution: Selective Reduced Integration (SRI). To resolve this, a Selective Reduced Integration scheme was implemented. The material stiffness matrix $\mathbf{C}$ was explicitly decomposed into normal (bending) and shear components:

$$\mathbf{C} = \mathbf{C}_{norm} + \mathbf{C}_{shear} = \begin{bmatrix} C_{11} & C_{12} & 0 \\ C_{21} & C_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & C_{33} \end{bmatrix} \quad (18)$$

The element stiffness matrix is then assembled using a mixed quadrature rule. We apply full integration (2×2) to the normal terms to maintain the element's rank and stability, while applying reduced integration (1×1) to the shear terms to eliminate the parasitic energy at the centroid:

$$\mathbf{K}_{el} = \underbrace{\sum_{i=1}^4 w_i (\mathbf{B}^T \mathbf{C}_{norm} \mathbf{B})_i |J|_i}_{\text{Full Int. (2x2) - Bending}} + \underbrace{\sum_{k=1}^1 w_k (\mathbf{B}^T \mathbf{C}_{shear} \mathbf{B})_k |J|_k}_{\text{Reduced Int. (1x1) - Shear}} \quad (19)$$

As shown in Table 2, this mathematical correction successfully removed the artificial stiffness, recovering the physical frequency of 7.10 Hz.

Table 2: Verification of Dry Structure Natural Frequency (Mode 1). Note the massive improvement after applying SRI.

Method	Frequency (Hz)	Error (%)
Analytical (Beam Theory)	7.08 Hz	-
FEM (Initial Full Integration)	103.08 Hz	1356.30 %
FEM (Corrected SRI)	7.10 Hz	0.28 %

4.2. Full Tank Frequency Response Analysis

With the elementary matrices validated, the full "sandwich" system (Left Fluid - Structure - Right Fluid) was analyzed.

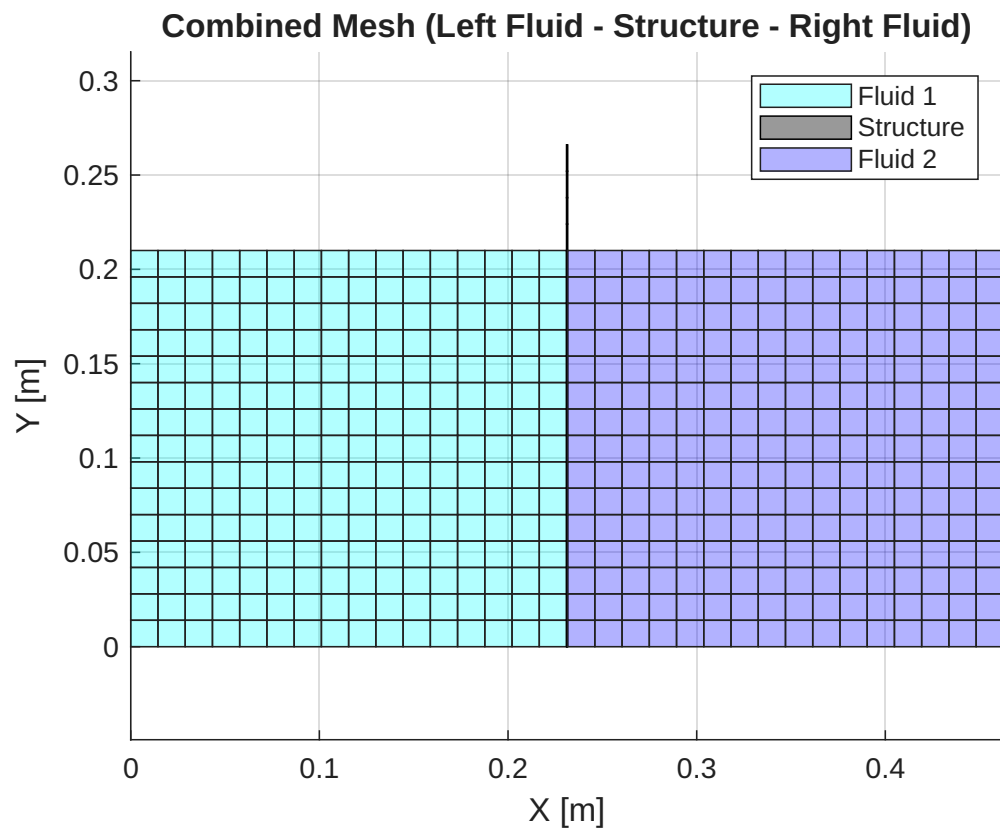


Figure 5: Combined Mesh (Left Fluid - Structure - Right Fluid). The structure is sandwiched between two identical fluid domains, creating the full hydroelastic coupling interface .

4.2.1. Modal Analysis

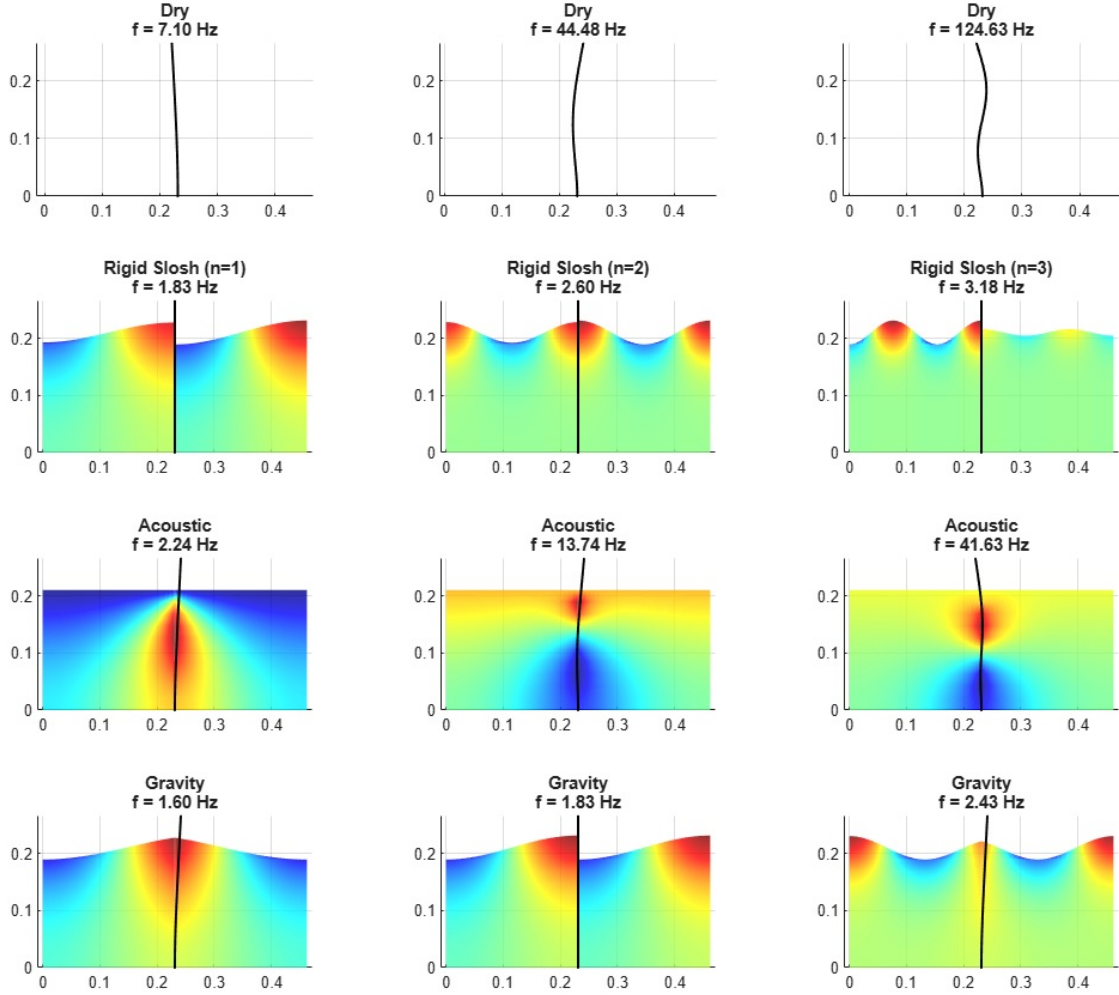


Figure 6: Modal Comparison Results: Dry Structure vs Rigid Sloshing vs Coupled Modes.

Analysis:

- **Dry Structure ($f = 7.10$ Hz):** Matches beam theory.
- **Acoustic Coupling ($f = 2.24$ Hz):** When gravity is neglected, the fluid acts as a pure Added Mass, dropping the frequency by $\approx 70\%$.
- **Full Gravity Coupling ($f = 1.60$ Hz):** With gravity, the interaction with low-frequency sloshing modes further reduces the fundamental system frequency.

4.3. Frequency Response Functions (FRF)

The Frequency Response Function (FRF) for a unit tip load is shown in Figure 7. The response separates into two regimes:

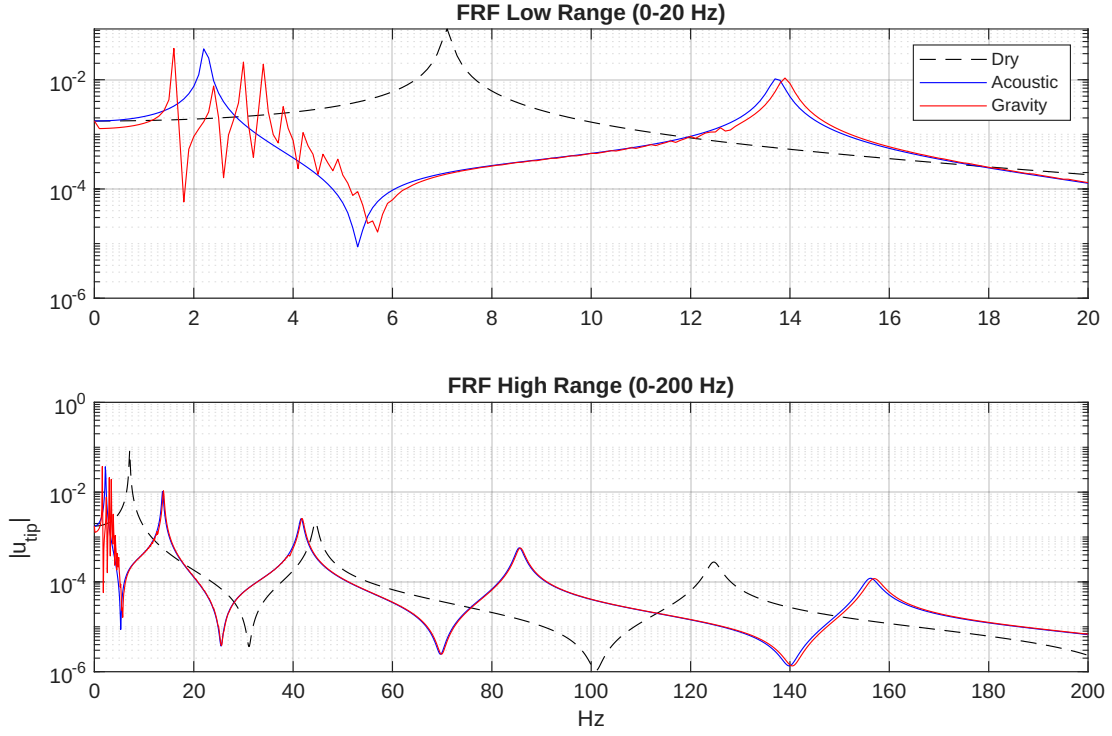


Figure 7: Frequency Response Function (FRF).

Regimes:

1. **Inertial Regime (> 10 Hz):** The "Acoustic" and "Gravity" curves are identical, confirming gravity is negligible for higher modes.
2. **Gravity Regime (< 10 Hz):** The curves diverge. The "Gravity" curve (Red) shows a distinct resonance peak at 1.60 Hz, compared to the "Acoustic" peak at 2.24 Hz.

4.4. Verification of Numerical Accuracy

To ensure reliability, the results were verified against analytical theories. The rigid sloshing modes matched the wave theory with near-zero error (Table 3).

Table 3: Verification of Rigid Sloshing Modes (FEM Quad4 vs. Analytical).

Mode (n)	Theory (Hz)	FEM (Hz)	Error (%)
1	1.8323	1.8323	0.00 %
2	2.5998	2.5999	0.01 %
3	3.1841	3.1847	0.02 %

5. Conclusion

This project involved the complete development of a Finite Element solver for hydroelastic problems. The study successfully validated the gravity stiffness implementation and resolved critical shear locking issues using Selective Reduced Integration.

5.1. Final Observations

- **Inertial Impact:** The fluid acts as a significant added mass, lowering the structural frequency by 70% (7.10 Hz → 2.24 Hz) even without gravity.
- **Gravity Coupling:** Gravity is essential for the low-frequency range (< 5 Hz), introducing a sloshing-structure interaction that shifts the resonance to 1.60 Hz.
- **Design Implication:** Standard acoustic solvers (neglecting gravity) are non-conservative for fundamental stability analysis in liquid storage systems.

References

- [1] C. Hoareau, "Linear Hydroelasticity," Course Notes, Cnam, 2025.
- [2] C. Hoareau, "Sloshing," Course Slides 2, Cnam, 2025.
- [3] Bathe, K. J. (2006). *Finite Element Procedures*. Prentice Hall.
- [4] Morand, H. J., & Ohayon, R. (1995). *Fluid Structure Interaction*. Wiley.
- [5] T. Irvine, "An Introduction to Fluid Slosh," Vibrationdata Tutorials, Rev 1.0, 2010.
- [6] Wikipedia contributors, "Dispersion (water waves)".